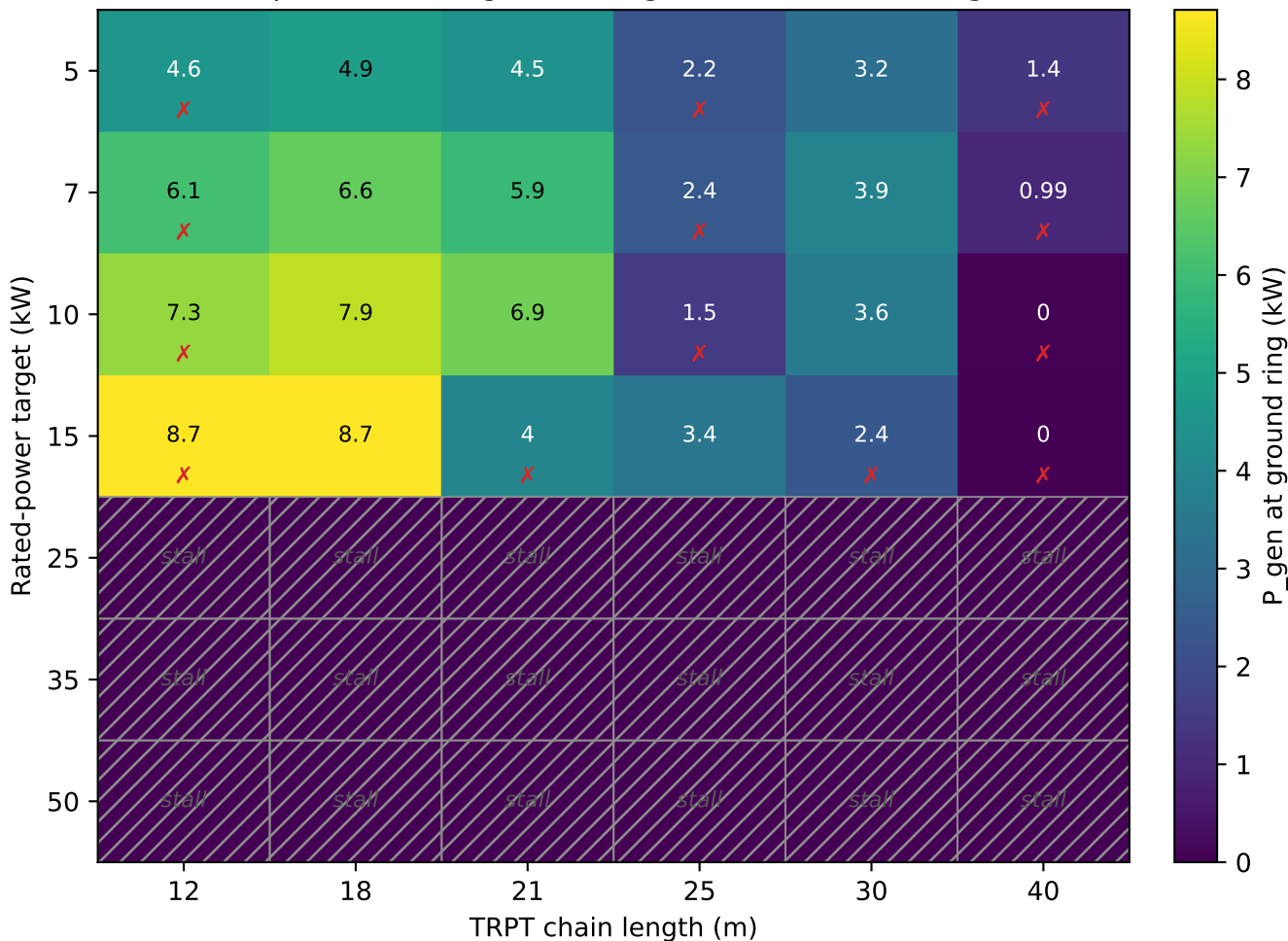


Delivered power at the ground ring — 42 baseline designs (v13)



Every cell is one simulation, not a test: the baseline design family scaled to a rated-power target (rows, 5–50 kW) and a chain length (columns, 12–40 m), run through the full ODE model for 30 s at 11 m/s; the number is the power delivered at the ground ring. The 5–15 kW targets deliver 4.6–8.7 kW at the short lengths (12–18 m) and degrade with length; the 40 m column is the twist wall — the longest chains cross their torsional limit (7–15 kW cells) and the 15 kW cell collapses to zero. The hatched 25–50 kW cells never spun up in the run — a gate start-up limitation at this scale, so they carry no verdict, not a failure. **X** = ran but failed the gate (power ≥ 2.5 kW, $\omega > 0.5$ rad/s, no twist crossing, ≥ 1.5 m clearance).